Cross-cultural recognition memory: intergroup correlation results

Bryan Medina — generated by `make_results_report.py`

June 3, 2026

Overview

Intergroup itemwise correlations and paired-bootstrap comparisons for two conditions (*Globalized-Music*, *Industrial-Nature*) and two trial types (hits and false alarms). All numbers were produced by the Python pipeline in `Analysis-Scripts-v2/python/`; see `STATS.md` / `stats.pdf` for the methods. Bootstrap settings: $n_{\text{boot}} = 2000$ for pairwise CIs, $n_{\text{boot}} = 5000$ for the paired test, attenuation correction in `fixed` mode, participant-level resampling, `minResp`$= 2$. Reported CIs are **percentile**: $\left[\mathrm{P}_{2.5}, \mathrm{P}_{97.5}\right]$ of the bootstrap distribution. Fisher-z back-transformed and BCa alternatives are shown in the diagnostic table beneath each headline. BCa was considered as the headline (it is the textbook recommendation for skewed bootstraps) but misfires in cells with low within-group reliability — for those cells the bias correction $\hat{z}_0$ is so large that $\hat{r}$ falls at or outside the BCa interval. The percentile CI is conservative under skew (wider than warranted) but doesn't break in that pathological way and is therefore the safer headline. The default p-value is the recentered-null variant.

Site partition and within-group reliabilities

Condition	Group	n_{subj}	n_{items}	$\hat{\rho}^{\text{SB}}_{\text{hit}}$	$\hat{\rho}^{\text{SB}}_{\text{fa}}$
Globalized-Music	US	96	80	0.776	0.865
Globalized-Music	SanBorja	63	80	0.716	0.741
Globalized-Music	Tsimane	68	80	0.217	0.695
Industrial-Nature	US	141	80	0.782	0.910
Industrial-Nature	SanBorja	56	80	0.707	0.802
Industrial-Nature	Tsimane	81	80	0.707	0.804
NHS	US	94	80	0.650	0.911
NHS	SanBorja	49	80	0.680	0.852
NHS	Tsimane	76	80	0.649	0.834

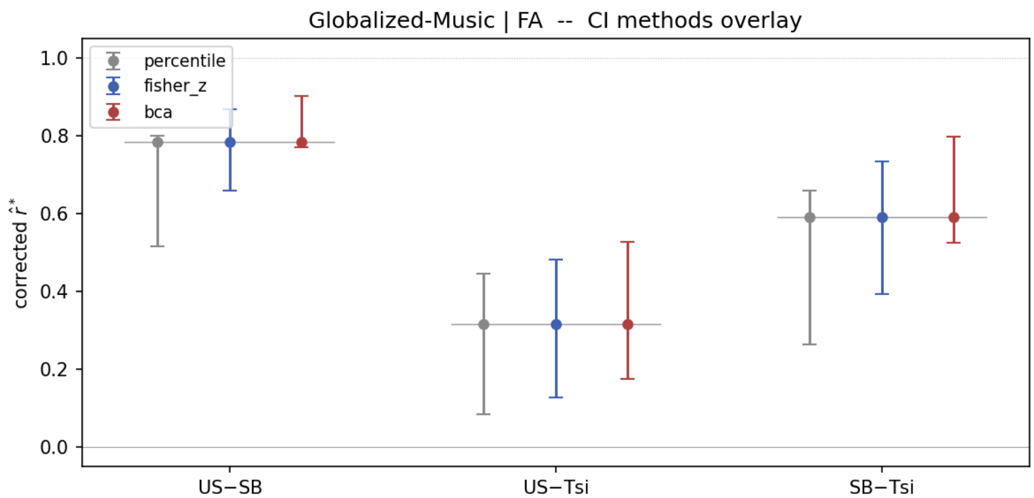
Globalized-Music

Hits

Pairwise intergroup correlations (percentile 95% CIs)

Pair	$\hat{r}$ (raw)	95% CI	$\hat{r}^*$ (corrected)	95% CI
US−SB	+0.478	[+0.236, +0.498]	+0.641	[+0.317, +0.669]
US−Tsi	+0.426	[+0.125, +0.425]	+1.038	[+0.306, +1.035]
SB−Tsi	+0.413	[+0.091, +0.442]	+1.047	[+0.230, +1.120]

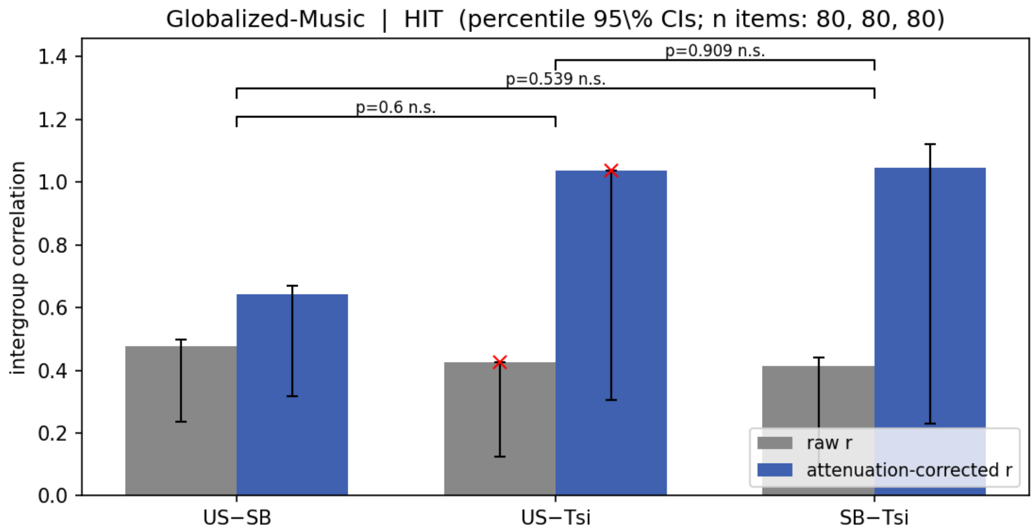
Diagnostic — alternative CIs on corrected $\hat{r}^*$



Pair	$\hat{r}^*$	Percentile 95% CI	Fisher-z 95% CI	BCa 95% CI
US−SB	+0.641	[+0.317, +0.669]	[+0.475, +0.763]	[+0.609, +0.805]
US−Tsi	+1.038	[+0.306, +1.035]	[+1.000, +1.000]	[+1.039, +1.299]
SB−Tsi	+1.047	[+0.230, +1.120]	[+0.998, +1.000]	[+0.961, +1.480]

Paired-bootstrap comparison

Comparison	Observed diff	95% CI (diff)	p (recentered)	p (straddle)
US-SB vs US-Tsi	+0.052	[-0.101, +0.291]	0.600	0.346
US-SB vs SB-Tsi	+0.065	[-0.115, +0.307]	0.539	0.384
US-Tsi vs SB-Tsi	+0.013	[-0.211, +0.210]	0.909	0.990

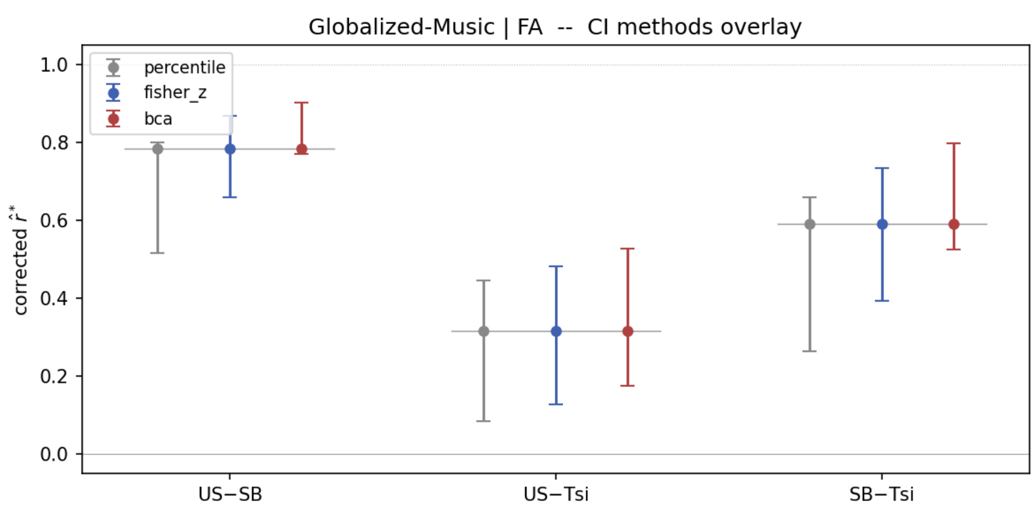


False alarms

Pairwise intergroup correlations (percentile 95% CIs)

Pair	$\hat{r}$ (raw)	95% CI	$\hat{r}^*$ (corrected)	95% CI
US−SB	+0.628	[+0.414, +0.640]	+0.784	[+0.517, +0.800]
US−Tsi	+0.245	[+0.065, +0.345]	+0.316	[+0.084, +0.446]
SB−Tsi	+0.424	[+0.190, +0.473]	+0.591	[+0.264, +0.659]

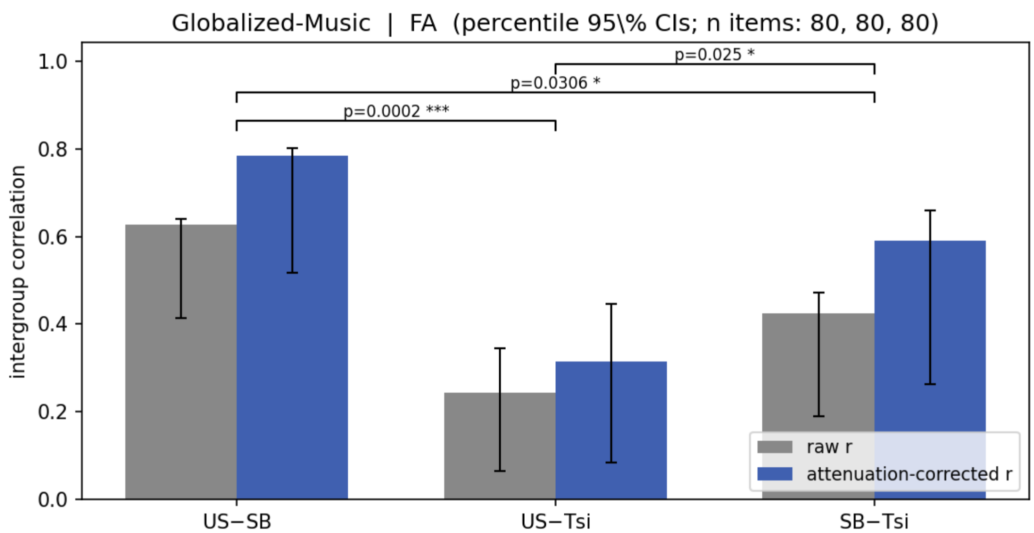
Diagnostic — alternative CIs on corrected $\hat{r}^*$



Pair	$\hat{r}^*$	Percentile 95% CI	Fisher-z 95% CI	BCa 95% CI
US−SB	+0.784	[+0.517, +0.800]	[+0.659, +0.867]	[+0.770, +0.902]
US−Tsi	+0.316	[+0.084, +0.446]	[+0.128, +0.482]	[+0.176, +0.529]
SB−Tsi	+0.591	[+0.264, +0.659]	[+0.394, +0.735]	[+0.525, +0.798]

Paired-bootstrap comparison

Comparison	Observed diff	95% CI (diff)	p (recentered)	p (straddle)
US-SB vs US-Tsi	+0.383	[+0.153, +0.493]	<0.001	<0.001
US-SB vs SB-Tsi	+0.204	[+0.008, +0.384]	0.031	0.040
US-Tsi vs SB-Tsi	-0.179	[-0.281, +0.028]	0.025	0.106



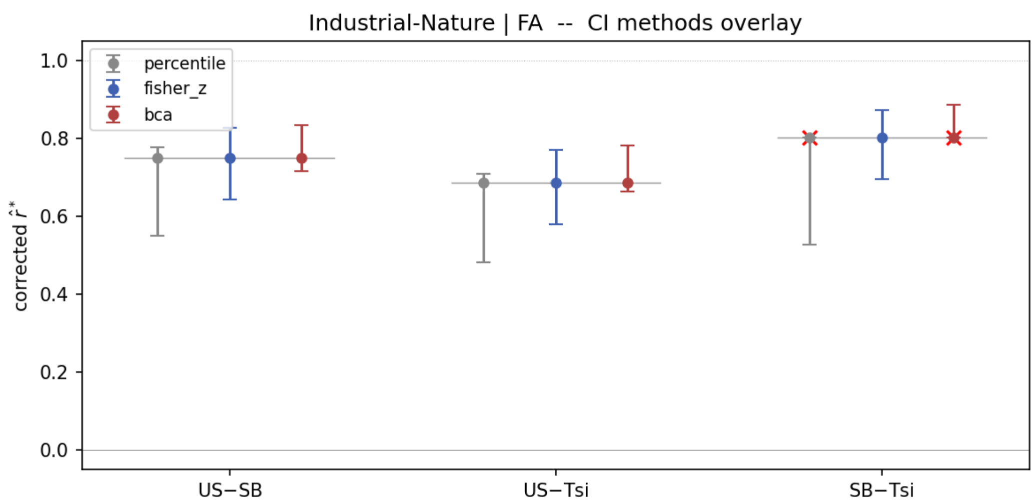
Industrial-Nature

Hits

Pairwise intergroup correlations (percentile 95% CIs)

Pair	$\hat{r}$ (raw)	95% CI	$\hat{r}^*$ (corrected)	95% CI
US−SB	+0.623	[+0.368, +0.608]	+0.838	[+0.496, +0.818]
US−Tsi	+0.570	[+0.304, +0.563]	+0.766	[+0.409, +0.757]
SB−Tsi	+0.722	[+0.428, +0.665]	+1.022	[+0.606, +0.940]

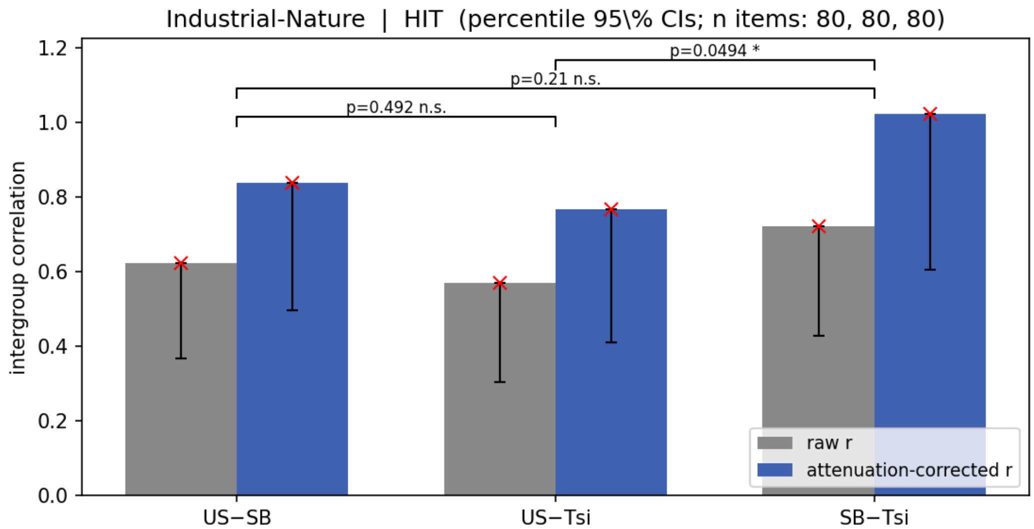
Diagnostic — alternative CIs on corrected $\hat{r}^*$



Pair	$\hat{r}^*$	Percentile 95% CI	Fisher-z 95% CI	BCa 95% CI
US−SB	+0.838	[+0.496, +0.818]	[+0.722, +0.909]	[+0.859, +0.893]
US−Tsi	+0.766	[+0.409, +0.757]	[+0.626, +0.859]	[+0.776, +0.874]
SB−Tsi	+1.022	[+0.606, +0.940]	[+1.000, +1.000]	[+1.025, +1.025]

Paired-bootstrap comparison

Comparison	Observed diff	95% CI (diff)	p (recentered)	p (straddle)
US-SB vs US-Tsi	+0.053	[-0.098, +0.211]	0.492	0.483
US-SB vs SB-Tsi	-0.099	[-0.208, +0.099]	0.210	0.482
US-Tsi vs SB-Tsi	-0.152	[-0.261, +0.043]	0.049	0.162

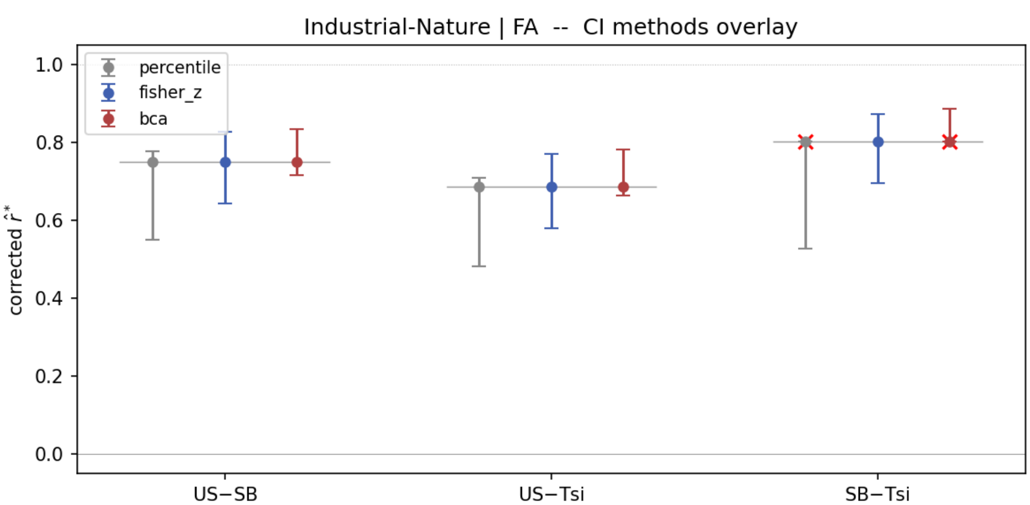


False alarms

Pairwise intergroup correlations (percentile 95% CIs)

Pair	$\hat{r}$ (raw)	95% CI	$\hat{r}^*$ (corrected)	95% CI
US−SB	+0.640	[+0.470, +0.665]	+0.750	[+0.550, +0.778]
US−Tsi	+0.588	[+0.413, +0.608]	+0.687	[+0.483, +0.710]
SB−Tsi	+0.644	[+0.425, +0.640]	+0.802	[+0.529, +0.797]

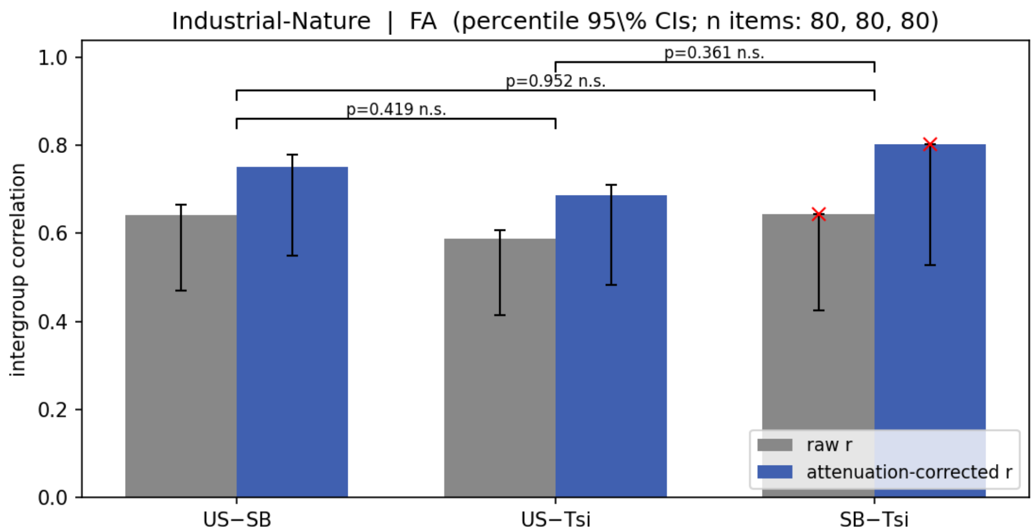
Diagnostic — alternative CIs on corrected $\hat{r}^*$



Pair	$\hat{r}^*$	Percentile 95% CI	Fisher-z 95% CI	BCa 95% CI
US−SB	+0.750	[+0.550, +0.778]	[+0.643, +0.827]	[+0.716, +0.835]
US−Tsi	+0.687	[+0.483, +0.710]	[+0.581, +0.771]	[+0.663, +0.781]
SB−Tsi	+0.802	[+0.529, +0.797]	[+0.695, +0.874]	[+0.804, +0.886]

Paired-bootstrap comparison

Comparison	Observed diff	95% CI (diff)	p (recentered)	p (straddle)
US-SB vs US-Tsi	+0.053	[-0.069, +0.189]	0.419	0.344
US-SB vs SB-Tsi	-0.004	[-0.088, +0.169]	0.952	0.540
US-Tsi vs SB-Tsi	-0.056	[-0.142, +0.099]	0.361	0.699



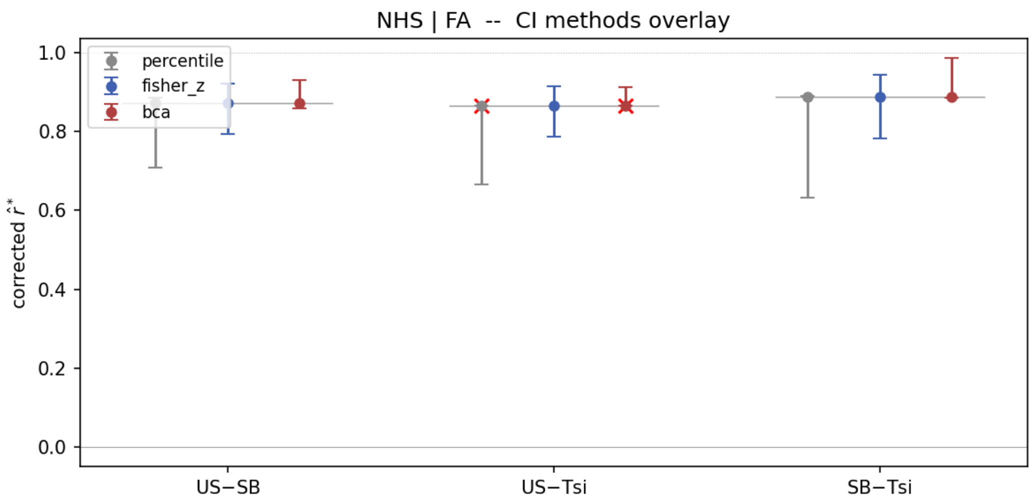
NHS

Hits

Pairwise intergroup correlations (percentile 95% CIs)

Pair	$\hat{r}$ (raw)	95% CI	$\hat{r}^*$ (corrected)	95% CI
US−SB	+0.491	[+0.255, +0.511]	+0.739	[+0.383, +0.769]
US−Tsi	+0.474	[+0.231, +0.502]	+0.730	[+0.356, +0.773]
SB−Tsi	+0.537	[+0.260, +0.542]	+0.808	[+0.391, +0.816]

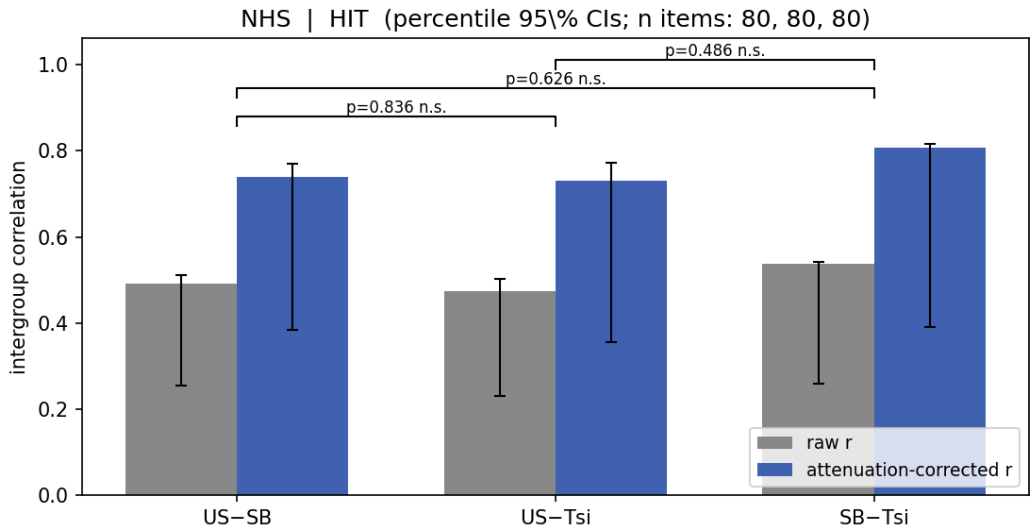
Diagnostic — alternative CIs on corrected $\hat{r}^*$



Pair	$\hat{r}^*$	Percentile 95% CI	Fisher-z 95% CI	BCa 95% CI
US−SB	+0.739	[+0.383, +0.769]	[+0.562, +0.851]	[+0.700, +0.865]
US−Tsi	+0.730	[+0.356, +0.773]	[+0.529, +0.853]	[+0.682, +0.962]
SB−Tsi	+0.808	[+0.391, +0.816]	[+0.632, +0.905]	[+0.797, +0.928]

Paired-bootstrap comparison

Comparison	Observed diff	95% CI (diff)	p (recentered)	p (straddle)
US-SB vs US-Tsi	+0.017	[-0.148, +0.182]	0.836	0.819
US-SB vs SB-Tsi	-0.046	[-0.204, +0.155]	0.626	0.802
US-Tsi vs SB-Tsi	-0.063	[-0.223, +0.132]	0.486	0.630

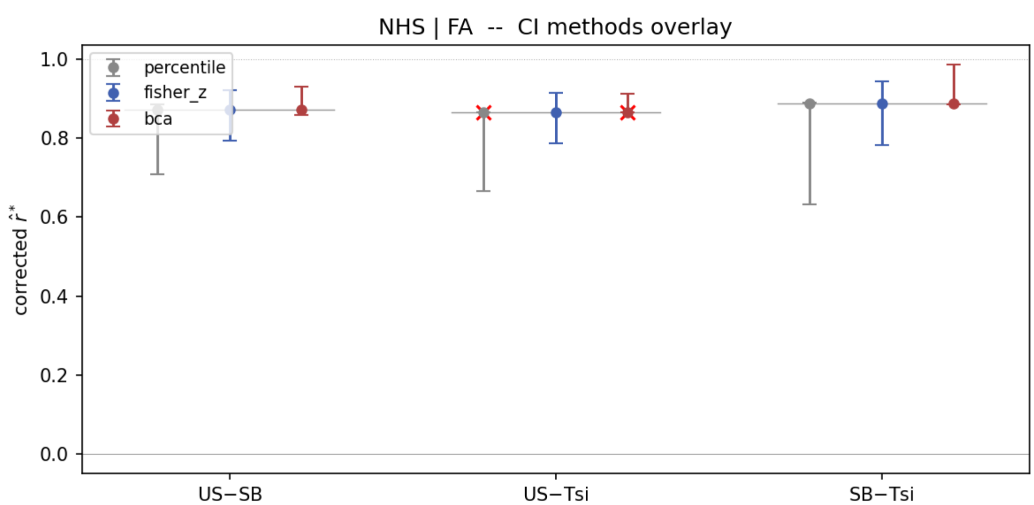


False alarms

Pairwise intergroup correlations (percentile 95% CIs)

Pair	$\hat{r}$ (raw)	95% CI	$\hat{r}^*$ (corrected)	95% CI
US−SB	+0.768	[+0.624, +0.780]	+0.872	[+0.708, +0.885]
US−Tsi	+0.753	[+0.580, +0.749]	+0.864	[+0.666, +0.860]
SB−Tsi	+0.748	[+0.533, +0.751]	+0.887	[+0.632, +0.891]

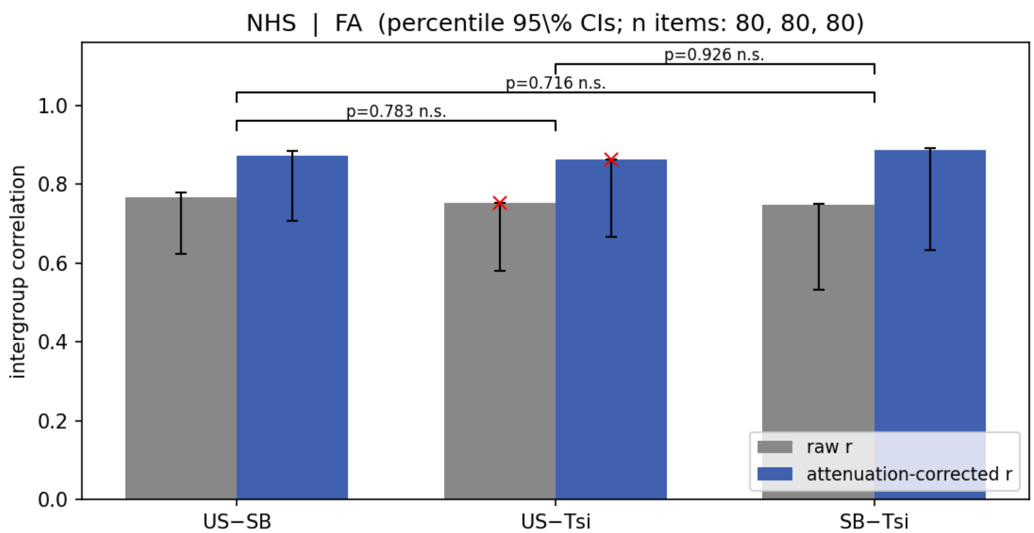
Diagnostic — alternative CIs on corrected $\hat{r}^*$



Pair	$\hat{r}^*$	Percentile 95% CI	Fisher-z 95% CI	BCa 95% CI
US−SB	+0.872	[+0.708, +0.885]	[+0.795, +0.921]	[+0.859, +0.930]
US−Tsi	+0.864	[+0.666, +0.860]	[+0.786, +0.915]	[+0.872, +0.913]
SB−Tsi	+0.887	[+0.632, +0.891]	[+0.783, +0.943]	[+0.884, +0.986]

Paired-bootstrap comparison

Comparison	Observed diff	95% CI (diff)	p (recentered)	p (straddle)
US-SB vs US-Tsi	+0.015	[-0.080, +0.149]	0.783	0.531
US-SB vs SB-Tsi	+0.020	[-0.053, +0.165]	0.716	0.350
US-Tsi vs SB-Tsi	+0.005	[-0.098, +0.138]	0.926	0.757



Methods note

All correlations are Spearman. Within-group reliabilities are split-half (Spearman–Brown corrected) computed by participant split. Attenuation correction applied as $\hat{r}^* = \hat{r}/\sqrt{\hat{\rho}_A^{\mathrm{SB}}\hat{\rho}_B^{\mathrm{SB}}}$; *not* clamped to $[-1, 1]$ (when within-group reliabilities are small relative to the raw correlation, $\hat{r}^*$ can legitimately exceed 1 — this is a property of the formula and a substantive observation about within-group noise, not a numerical failure). Bootstrap CIs computed three ways from the same resampled distribution: **percentile** (the $[\mathrm{P}_{2.5}, \mathrm{P}_{97.5}]$ quantiles), **Fisher-z back-transformed** (normal-approx CI on the z-scale, using bootstrap z-space SD, centered on $\operatorname{atanh}(\hat{r})$), and **BCa** (bias-corrected and accelerated with jackknife). The headline is percentile; BCa is shown in the diagnostic table. We chose percentile as the headline after observing that BCa misfires in cells with low within-group reliability (Tsimane' Globalized-Music) — there the bias correction $\hat{z}_0$ becomes extreme and the corrected interval excludes the sample point estimate. Percentile is conservative under skew (wider than warranted) but doesn't break in that pathological way. Paired-bootstrap comparisons share the resample of any group that appears in both correlations under test; p-values are reported under two definitions, recentered-null (recommended) and straddle-zero (legacy).